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Critical Thinking and Creativity Skills

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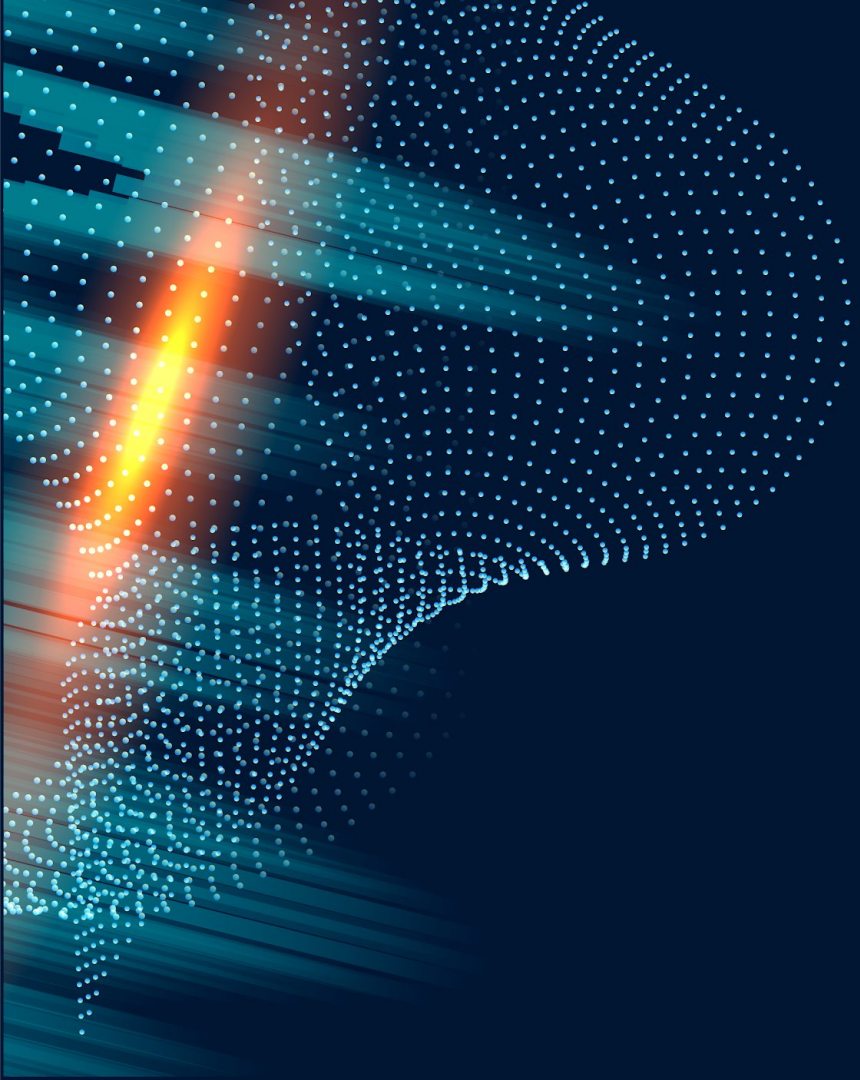

Various Problem-Solving Methodologies

“The problem is not that there are problems. The problem is expecting otherwise and thinking that having problems is a problem.”

—Theodore Rubin

From Analysis to Solution

Analytical Thinking gave us the map.

- It helped us understand the terrain of the problem.
- We can identify the components, the relationships, and the flaws in logic.

Problem-Solving Methodologies give us the navigation system.

- They provide a structured process for moving from the problem state to the solution state.
- They prevent us from just driving in circles or heading in a random direction.

Methodology 1: The Scientific Method

A Rigorous, Evidence-Based Approach

1. **Observe & Question**: A system is behaving anomalously.
2. **Form a Hypothesis**: "I hypothesize that the slowdown is caused by a memory leak in Service X."
3. **Make a Prediction**: "If Service X is restarted, memory usage will stabilize and performance will return to normal."

Methodology 1: The Scientific Method

A Rigorous, Evidence-Based Approach

4. **Test (Experiment)**: Restart Service X in a controlled, monitored environment.

5. **Analyze Results**: Did memory usage stabilize? Did performance improve?

6. **Iterate**: If the hypothesis was wrong, form a new one ("Perhaps it's a database lock...").

Methodology 2: Divide and Conquer

Taming Complexity by Decomposition

- **Principle:** Break a large, complex problem into smaller, more manageable sub-problems.
- **Solve** each sub-problem independently.
- **Combine** the solutions to solve the original problem.

Methodology 3: Lateral Thinking

Thinking "Outside the Box"

- **Coined by Edward de Bono.**
- **It's about changing concepts, perceptions, and moving sideways to generate new ideas, rather than moving forward with traditional step-by-step logic.**

Methodology 3: Lateral Thinking

Key Technique: Provocation

- **Challenge assumptions. Ask "What if...?" even if it seems silly.**
  - **"What if the attacker wants us to see their malware?"**
  - **"What if the most secure system is the one we think is already compromised?"**
  - **"What if we reward users for finding phishing links instead of punishing them for clicking?"**

Synthesis: The Art of Choosing

No single methodology is a silver bullet.

- **The expert's skill is in selecting and blending them.**

- **Your thought process should be fluid:**
- **Example of an Attack :**
  - **Divide the problem of the data breach.**
  - **Form a scientific hypothesis about the exfiltration method.**
  - **Use lateral thinking to brainstorm how an insider might hide their activity.**

THANKS!

Do you have any questions?

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